Functions

Chapter 1

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Outline of Lecture

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4.1 Introduction

Definition 4.1.1: Definition of a function

A function from a set A to a set B is a rule that assigns to each element x in set A exactly one element $y = f(x)$ as shown in Figure 1.

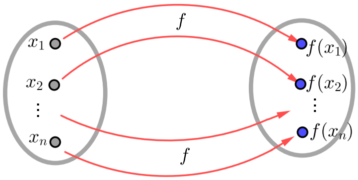

Figure 1: Definition of a function

1. $y = f(x)$ is called the **image** of x under f while x is called the **pre-image** of y under f.

2. The sets A and B are called **Domain** and **Codomain** of the function f respectively; often denoted by D_f and C_f.

3. The set of all possible values of f as x varies throughout the domain is called the **range** of the function f, often denoted by R_f and it is given by

$$R_f = \{f(x)|x \in A)\} \tag{1}$$

Since $y = f(x)$, then x is the *independent* variable and y is the *dependent* variable.

Example: Consider the function given by

$$A(r) = \pi r^2$$

<u>Solution</u>: The above function gives the area of a circle as a function of its radius. The area of a circle of radius 2 is $A(2) = 4\pi$.

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Limit at a point

To introduce the concept of limits let's investigate the behavior of the function defined by

$$f(x) = \frac{x^3 - 8}{x^2 - 4}$$

for values of near 2. This function is not defined at $x = 2$. The following table gives values of $f(x)$ for values of x close to 2, but not equal to 2.

x	1.9	1.99	1.999	2	2.001	2.01	2.1
$f(x)$	2.926	2.993	2.999	*	3.001	3.008	3.076
$\lim_{x\to 2} f(x)$			$\longrightarrow$	*	$\longleftarrow$		

Table 1: Guessing $\lim_{x\to 2} f(x)$.

From Table 1 we see that when x is close to 2 (on either side of 2), $f(x)$ is close to 3. We express this by saying "the limit of the function $f(x) = \frac{x^3-8}{x^2-4}$ as x approaches 2 is equal to 3." Denoted by

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{x^3 - 8}{x^2 - 4} = 3$$

Definition 4.8.1 Definition of limit

We write

$$\lim_{x \to a} f(x) = L \tag{2}$$

and say that "the limit of $f(x)$, as x approaches a, equals to L.

Figure 2 shows the graphs of the three functions where in each case, $\lim_{x\to a} f(x) = L$.

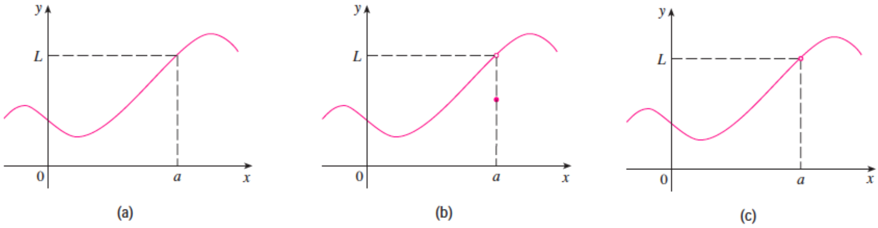

Figure 2: $\lim_{x\to a} f(x) = L$ in all three cases

From Figure 2 we observe that in (a), $f(a) = L$, in (b), $f(a) \neq L$ and in (c), $f(a)$ is not defined. However, in each case, it is true that $\lim_{x\to a} f(x) = L$.

Example: Guess the values of the following limits

a) $\lim_{x\to 1} \frac{x-1}{x^2-1}$

b) $\lim_{x\to 1} g(x)$ where $g(x) = \begin{cases} \frac{x-1}{x^2-1}, & \text{if} \quad x \neq 1 \\ 2, \quad \text{if} & x = 1 \end{cases}$

c) $\lim_{t\to 0} \frac{\sqrt{t^2+9}-3}{t^2}$

Solution:

a) The function $f(x) = \frac{x-1}{x^2-1}$ is not defined when $x = 1$, we will use the table to compute values of $f(x)$ for values of x close to 1, but not equal to 1 as shown below.

x	0.99	0.999	0.9999	1	1.0001	1.001	1.01
$f(x)$	0.502513	0.500250	0.500025	*	0.499975	0.499750	0.497512
			$\longrightarrow$	*	$\longleftarrow$		

Table 2: Guessing $\lim_{x\to 1} f(x)$

From Table 2, we observe that $f(x) \to 0.5$ from either side as $x \to 1$. So, we can make an educated guess that

$$\lim_{x \to 1} \frac{x-1}{x^2-1} = 0.5$$

□

- b) The function $g(x)$ resulted from the function $f(x)$ in (a) by assigning the value 2 when $x = 1$. So, the limit of $f(x)$ and $g(x)$ as x approaches 1 are equal as shown in Figures 3 and 4 □

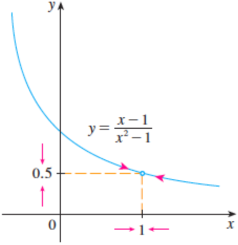

Figure 3

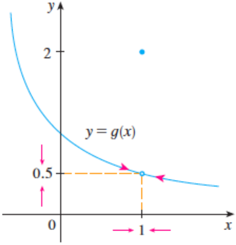

Figure 4

- Table 3 below shows the list of values of the function $f(t) = \frac{\sqrt{t^2+9}-3}{t^2}$ for several values of t near 0.

t	-0.5	-0.1	-0.05	0.01	0	0.01	0.05	0.1	0.5
$f(t)$	0.16662	0.16662	0.16666	0.16667	*	0.16667	0.16666	0.16662	0.16662
				$\longrightarrow$	*	$\longleftarrow$			

Table 3: Guessing $\lim_{t\to 0} f(t)$

From Table 3 as t approaches 0, the values of the function seem to approach 0.1666666... and so we make an educated guess that

$$\lim_{t\to 0} \frac{\sqrt{t^2+9}-3}{t^2} = \frac{1}{6}$$

□

Definition 4.8.2 One-sided limits

A one-sided limit can be the **left-hand limit** which is the limit as $f(x)$ approaches a from the left written as

$$\lim_{x \to a^-} f(x) = L \tag{3}$$

or the **right-hand limit** which is the limit as $f(x)$ approaches a from the right written as

$$\lim_{x \to a^+} f(x) = L \tag{4}$$

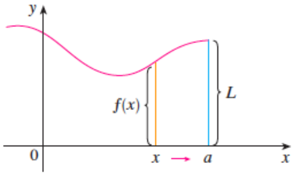

Figure 5: $\lim_{x \to a^-} f(x) = L$

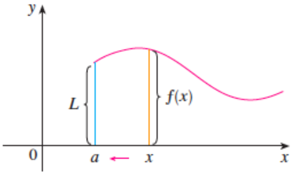

Figure 6: $\lim_{x \to a^+} f(x) = L$

Definition 5.8.3 Existence of Limits

For the function $f(x)$ the limit exists and it is given by

$$\lim_{x \to a} f(x) = L$$

if and only if

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

Example: Find $\lim_{x \to 1} f(x)$ for the function

$$f(x) = \begin{cases} 1 - 2x, & x < 1 \\ x - 3, & x \geq 1 \end{cases}$$

Solution: The graph of $f(x)$ is shown in Figure 7 below.

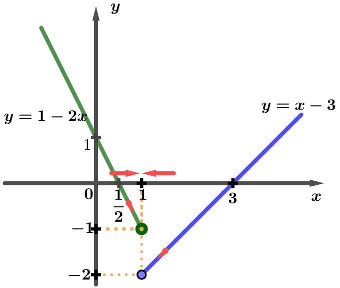

Figure 7: Graph of $f(x)$

We have that

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} (1 - 2x) = 1 - 2(1) = -1$$

and

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1} (x - 3) = 1 - 3 = -2$$

Since

$$\lim_{x\to 1^-} f(x) = -1 \neq -2 = \lim_{x\to 1^+} f(x)$$

then $\lim_{x\to 1} f(x)$ does not exist.

$\square$

Theorem 4.8.1 Limit Laws

Let $f(x)$ and $g(x)$ be functions such that

$$\lim_{x\to a} f(x) = L_1 \quad \text{and} \quad \lim_{x\to a} g(x) = L_2$$

and c be a constant. Then

1. $\lim_{x\to a}[f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = L_1 + L_2$
2. $\lim_{x\to a}[f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x) = L_1 - L_2$
3. $\lim_{x\to a}[cf(x)] = c\lim_{x\to a} f(x) = cL_1$
4. $\lim_{x\to a}[f(x)g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x) = L_1 L_2$

6. $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} = \frac{L_1}{L_2}$, if $\lim_{x\to a} g(x) = L_2 \neq 0$
7. $\lim_{x\to a} [f(x)]^n = [\lim_{x\to a} f(x)]^n$, where n is a positive integer
8. $\lim_{x\to a} c = c$
9. $\lim_{x\to a} x = a$
10. $\lim_{x\to a} x^n = a^n$, where n is a positive integer
11. $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$, where n is a positive integer (If n is even, we assume that $a > 0$).
12. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$, where n is a positive integer (If n is even, we assume that $\lim_{x\to a} f(x) > 0$).

Example: Find the following limits

a) $\lim_{x\to 2} x^2$
b) $\lim_{x\to 3}(x^2 + 5)$
c) $\lim_{x\to 3} 4x^2$
d) $\lim_{x\to 5}(2x^2 - 3x + 4)$
e) $\lim_{x\to -2} \frac{x^3+2x^2-1}{5-3x}$

Solution:

a) $\lim_{x\to 2} x^2 = \lim_{x\to 2} c \cdot c = \lim_{x\to 2} \cdot \lim_{x\to 2} x = 2 \cdot 2 = 4$
b) $\lim_{x\to 3}(x^2 + 5) = \lim_{x\to 3} x^2 + \lim_{x\to 3} 5 = 3^2 + 5 = 14$
c) $\lim_{x\to 3} 4x^2 = 4 \lim_{x\to 3} x^2 = 4 \cdot 3^2 = 36$

d)

$$\begin{aligned}\lim_{x\to 5}(2x^2-3x+4) &= \lim_{x\to 5}(2x^2)-\lim_{x\to 5}(3x)+\lim_{x\to 5}4\\ &= 2\lim_{x\to 5}x^2-3\lim_{x\to 5}x+\lim_{x\to 5}4\\ &= 2(5^2)-3(5)+4\\ &= 39\end{aligned}$$

e)

$$\begin{aligned}\lim_{x\to -2}\frac{x^3+2x^2-1}{5-3x} &= \frac{\lim_{x\to -2}(x^3+2x^2-1)}{\lim_{x\to -2}(5-3x)}\\ &= \frac{\lim_{x\to -2}x^3+2\lim_{x\to -2}x^2-\lim_{x\to -2}1}{\lim_{x\to -2}5-3\lim_{x\to -2}x}\\ &= \frac{(-2)^3+2(-2)^2-1}{5-3(-2)}\\ &= -\frac{1}{11}\end{aligned}$$

Theorem 4.8.2

If $f(x) = a_n x^n + a_{n-1}x^{n-1}a_{n-2}x^{n-2} + \cdots + a_0$ is a polynomial function, and c is any number, then
$\lim_{x\to c} f(x) = f(c) = a_n c^n + a_{n-1}c^{n-1} + a_{n-2}c^{n-2} + \cdots + a_0$.

Theorem 4.8.3

If $f(x)$ and $g(x)$ are polynomials, and c is a constant, then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{f(c)}{g(c)}, \qquad \text{provided that } g(c) \neq 0$$

Example: Determine the following limits

a) $\lim_{x\to 3} x^2(2-x)$

b) $\lim_{x\to 2} \frac{x^3+4x^2-3}{x^2+5}$

c) $\lim_{x\to -5} \frac{x^2-25}{3(x+5)}$

<u>*Solution*</u>:

a) Since $f(x) = x^2(2-x) = -x^3 + 2x^2$ is a polynomial, then

$$\lim_{x \to 3} f(x) = f(3) = -(3)^3 + 2(3^2) = -9$$

b) We have that $f(x) = x^3 + 4x^2 - 3$ and $g(x) = x^2 + 5$. So, $f(2) = 2^3 + 4(2^2) - 3 = 21$ and $g(2) = 2^2 + 5 = 9$. Thus,

$$\lim_{x \to 2} \frac{f(x)}{g(x)} = \frac{f(2)}{g(2)} = \frac{21}{9} = \frac{7}{3}$$

c) We have that $f(x) = x^2 - 25$ and $g(x) = 3(x+5)$. So $f(-5) = (-5)^2 - 25$ and $g(-5) = 3(-5+5) = 0$, then $\frac{f(-5)}{g(-5)}$ is undefined. Hence

$$\lim_{x \to -5} \frac{f(x)}{g(x)} = \lim_{x \to -5} \frac{(x-5)\cancel{(x+5)}}{3\cancel{(x+5)}} = \lim_{x \to -5} \frac{x-5}{3} = \frac{-10}{3}$$

4.8.1 Trigonometric Functions and Square Roots

Consider the function

$$f(x) = \frac{\sin x}{x}$$

To guess

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin x}{x}$$

We consider the following table

x	-0.01	-0.005	-0.001	0	0.001	0.05	0.01
$f(x)$	0.9999833	0.9999958	0.9999998	*	0.9999998	0.9999958	0.9999833
			$\longrightarrow$	*	$\longleftarrow$		

Table 4: Guessing $\lim_{x \to 0} f(x)$

From Table 3 as x approaches 0, the values of the function seem to approach 1. Therefore

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \tag{5}$$

We use Equation 5 to find the limit as x approaches 0 of trigonometric functions of similar form.

Example: Determine the following limits

a) $\lim_{x \to 0} \frac{\sin(3x)}{4x}$

b) $\lim_{x \to 0} 2x \cot x$

c) $\lim_{x \to 0} \frac{\sin \theta}{2\theta}$

d) $\lim_{x \to 0} \frac{3x - \tan(2x)}{2x}$

Solution:

a) Since $x \to 0$, then $3x \to 0$. Thus

$$\begin{aligned}
\lim_{x\to 0} \frac{\sin(3x)}{4x} &= \lim_{x\to 0} \frac{\sin(3x)}{4x \cdot \frac{3}{3}} \\
&= \lim_{x\to 0} \frac{3}{4} \frac{\sin(3x)}{3x} \\
&= \frac{3}{4} \lim_{x\to 0} \frac{\sin(3x)}{3x} \\
&= \frac{3}{4} \cdot 1 \\
&= \frac{3}{4}
\end{aligned}$$

b) Note that $\cot x = \frac{\cos x}{\sin x}$ and $2x \to 0$ as $x \to 0$. Thus

$$\begin{aligned}
\lim_{x\to 0} 2x\cot x &= \lim_{x\to 0} 2x\cdot\frac{\cos x}{\sin x}\\
&= 2\lim_{x\to 0}\frac{x}{\sin x}\cdot\cos x\\
&= 2\lim_{x\to 0}\frac{1}{\frac{\sin x}{x}}\cdot\cos x\\
&= 2\frac{\lim_{x\to 0} 1}{\lim_{x\to 0}\frac{\sin x}{x}}\cdot\lim_{x\to 0}\cos x\\
&= 2\cdot\frac{1}{1}\cdot 1\\
&= 2
\end{aligned}$$

c) $\lim_{\theta\to 0}\frac{\sin\theta}{2\theta} = \lim_{\theta\to 0}\frac{\frac{\sin\theta}{\theta}}{\frac{\sin 2\theta}{\theta}}$. Thus

$$\lim_{\theta\to 0} \frac{\frac{\sin\theta}{\theta}}{\frac{\sin 2\theta}{\theta}} = \frac{\lim_{\theta\to 0} \frac{\sin\theta}{\theta}}{\lim_{\theta\to 0} \frac{\sin 2\theta}{\theta}}$$

$$= \frac{\lim_{\theta\to 0} \frac{\sin\theta}{\theta}}{\lim_{\theta\to 0} \frac{\sin 2\theta}{\theta\cdot\frac{2}{2}}}$$

$$= \frac{\lim_{\theta\to 0} \frac{\sin\theta}{\theta}}{2\lim_{\theta\to 0} \frac{\sin 2\theta}{2\theta}}$$

$$= \frac{1}{2}$$

$\therefore \lim_{\theta\to 0} \frac{\sin\theta}{2\theta} = \frac{1}{2}$

d) $\lim_{x\to 0} \frac{3x-\tan(2x)}{2x} = \lim_{x\to 0} \left(\frac{3}{2} - \frac{\sin 2x}{2x\cos 2x}\right)$

$$
\begin{aligned}
\lim_{x\to 0}\left(\frac{3}{2}-\frac{\sin 2x}{2x\cos 2x}\right) &= \lim_{x\to 0}\left(\frac{3}{2}-\frac{\sin 2x}{2x}\cdot\frac{1}{\cos 2x}\right)\\
&= \lim_{x\to 0}\frac{3}{2}-\lim_{x\to 0}\frac{\sin 2x}{2x}\cdot\lim_{x\to 0}\frac{1}{\cos x}\\
&= \frac{3}{2}-1\cdot 1=\frac{1}{2}
\end{aligned}
$$

Note that for the function $f(x)=\frac{\sqrt{t^2+9}-3}{t^2}$ in Example 4.14 (c) we made an educated guess that

$$
\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}=\frac{1}{6}
$$

However, this limit can also be computed algebraically as follows

$$\lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2} = \lim_{t\to 0}\frac{\sqrt{t^2+9}-3}{t^2}\cdot\frac{\sqrt{t^2+9}+3}{\sqrt{t^2+9}+3}$$

$$= \lim_{t\to 0}\frac{t^2+9-9}{t^2(\sqrt{t^2+9}+3)}$$

$$= \lim_{t\to 0}\frac{\cancel{t^2}}{\cancel{t^2}(\sqrt{t^2+9}+3)}$$

$$= \lim_{t\to 0}\frac{1}{\sqrt{t^2+9}+3} = \frac{1}{6}$$

Example: Compute the following limits

a) $\lim_{x\to 0}\frac{\sqrt{2x+7}-\sqrt{7}}{x}$

b) $\lim_{x\to 0}\frac{\sqrt{3x+1}-1}{x}$

c) $\lim_{h\to 0}\frac{\sqrt{2+h}-\sqrt{2}}{h}$

a)

$$\lim_{x\to 0}\frac{\sqrt{2x+7}-\sqrt{7}}{x}=\lim_{x\to 0}\frac{\sqrt{2x+7}-\sqrt{7}}{x}\cdot\frac{\sqrt{2x+7}+\sqrt{7}}{\sqrt{2x+7}+\sqrt{7}}$$
$$=\lim_{x\to 0}\frac{2x+7-7}{x(\sqrt{2x+7}+\sqrt{7})}$$
$$=\lim_{x\to 3}\frac{2\cancel{x}}{\cancel{x}(\sqrt{2x+7}+\sqrt{7})}$$
$$=\lim_{x\to 0}\frac{2}{\sqrt{2x+7}+\sqrt{7}}=\frac{1}{\sqrt{7}}$$

b)

$$\lim_{x\to 0}\frac{\sqrt{3x+1}-1}{x}=\lim_{x\to 0}\frac{\sqrt{3x+1}-1}{x}\cdot\frac{\sqrt{3x+1}+1}{\sqrt{3x+1}+1}$$
$$=\lim_{x\to 0}\frac{3x+1-1}{x(\sqrt{3x+1}+1)}=\lim_{x\to 3}\frac{3\cancel{x}}{\cancel{x}(\sqrt{3x+1}+1)}$$
$$=\lim_{x\to 0}\frac{3}{\sqrt{3x+1}+1}=\frac{3}{2}$$

c)

$$\lim_{h\to 0}\frac{\sqrt{2+h}-\sqrt{2}}{h} = \lim_{h\to 0}\frac{\sqrt{2+h}-\sqrt{2}}{h}\cdot\frac{\sqrt{2+h}+\sqrt{2}}{\sqrt{2x+7}+\sqrt{7}}$$
$$= \lim_{h\to 0}\frac{2+h-2}{h(\sqrt{2+h}+\sqrt{2})}$$
$$= \lim_{h\to 0}\frac{\cancel{h}}{\cancel{h}(\sqrt{2+h}+\sqrt{2})}$$
$$= \lim_{h\to 0}\frac{2}{\sqrt{2+h}+\sqrt{2}}$$
$$= \frac{2}{2\sqrt{2}}$$
$$= \frac{1}{\sqrt{2}}$$

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4.8.2 Infinite Limits

Consider the function

$$f(x) = \frac{1}{x^2}$$

To find

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{1}{x^2}$$

Let's consider the following table of values of $f(x)$ for values of x close to 0, but not equal to 0.

x	-0.05	-0.01	-0.001	0	0.001	0.01	0.05
$f(x)$	400	10,000	1,000,000	*	1,000,000	10,000	400
			$\longrightarrow$	*	$\longleftarrow$		

Table 5: Guessing $\lim_{x \to 0} f(x)$

As shown in Figure 8 and Table 5 the values of $f(x)$ can be made arbitrarily large by taking x values close enough to 0. The values of $f(x)$ do not approach a number, so $\lim_{x\to 0} \frac{1}{x^2}$ does not exist.

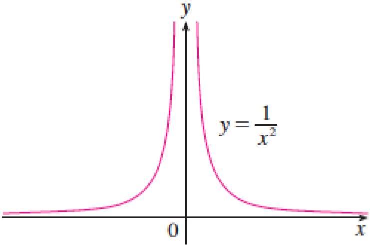

Figure 8: $\lim_{x\to 0} \frac{1}{x^2}$

To indicate this kind of behavior, we use the notation

$$\lim_{x\to 0} \frac{1}{x^2} = \infty$$

Definition 4.8.4

Let $f(x)$ be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty \tag{6}$$

This means the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a, but not equal to a.

Definition 4.8.5

Let $f(x)$ be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = -\infty \tag{7}$$

This means the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.

Similar definitions can be given for the one-sided infinite limits

$$\lim_{x \to a^-} f(x) = \infty, \quad \lim_{x \to a^+} f(x) = \infty$$
$$\lim_{x \to a^-} f(x) = -\infty, \quad \lim_{x \to a^+} f(x) = -\infty$$

as demonstrated on Figure 9 below

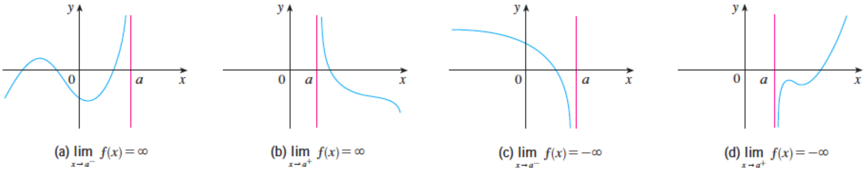

Figure 9: One-sided infinite limits

Definition 4.8.6

The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x\to a} f(x) = \infty, \quad \lim_{x\to a^-} f(x) = \infty, \quad \lim_{x\to a^+} f(x) = \infty$$
$$\lim_{x\to a} f(x) = -\infty \quad \lim_{x\to a^-} f(x) = -\infty, \quad \lim_{x\to a^+} f(x) = -\infty$$

Example: Determine the following limits

a) $f(x) = \frac{1}{x}$
  i) $\lim_{x\to 0^-} f(x)$
  ii) $\lim_{x\to 0^+} f(x)$

b) $g(x) = \frac{2x}{x-3}$
  i) $\lim_{x\to 3^-} g(x)$
  ii) $\lim_{x\to 3^+} g(x)$

Solution

a)
i) $f(x) = \frac{1}{x}$ increases in a negative direction as x approaches 0 from the left. Thus

$$\lim_{x \to 0^-} f(x) = -\infty$$

ii) $f(x) = \frac{1}{x}$ increases as x approaches 0 from the right. Thus

$$\lim_{x \to 0^-} f(x) = +\infty$$

b)
i) $g(x) = \frac{2x}{x-3}$ increases in a negative direction as x approaches 3 from the left. Thus

$$\lim_{x \to 3^-} f(x) = -\infty$$

ii) $g(x) = \frac{2x}{x-3}$ increases as x approaches 3 from the right. Thus

$$\lim_{x \to 3^+} f(x) = +\infty$$

4.8.3 Limits to Infinity

In this section, we want to look at the limit of the function $f(x)$ as $x \to \infty$ or $x \to -\infty$.

Consider the function

$$f(x) = \frac{1}{x}$$

As shown in Figure 10 below.

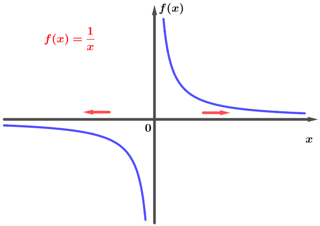

Figure 10: $\lim_{x \to \pm\infty} \frac{1}{x}$

This function is defined for all real numbers except at $x = 0$. Thus

a) When $x \to \infty$, $\frac{1}{x} \to 0^+$.
b) When $x \to -\infty$, $\frac{1}{x} \to 0^-$.

NB: If $f(x)$ is a constant function. i.e. $f(x) = k$, then $\lim_{x\to\infty} f(x) = k = \lim_{x\to-\infty} f(x)$

Theorem 4.8.4 Laws of limits to infinity

Let $f(x)$ and $g(x)$ be functions such that

$$\lim_{x\to\infty} f(x) = L_1 \quad \text{and} \quad \lim_{x\to\infty} g(x) = L_2, \quad \text{where } L_1, L_2 \in \mathbb{R}$$

and c be a constant. Then

1. $\lim_{x\to\infty}[f(x)+g(x)] = \lim_{x\to\infty} f(x) + \lim_{x\to\infty} g(x) = L_1 + L_2$
2. $\lim_{x\to\infty}[f(x)-g(x)] = \lim_{x\to\infty} f(x) - \lim_{x\to\infty} g(x) = L_1 - L_2$
3. $\lim_{x\to\infty}[cf(x)] = c\lim_{x\to\infty} f(x) = cL_1$
4. $\lim_{x\to\infty}[f(x)g(x)] = \lim_{x\to\infty} f(x)\cdot\lim_{x\to\infty} g(x) = L_1L_2$
5. $\lim_{x\to\infty}\frac{f(x)}{g(x)} = \frac{\lim_{x\to\infty} f(x)}{\lim_{x\to\infty} g(x)} = \frac{L_1}{L_2}$, if $\lim_{x\to\infty} g(x) = L_2 \neq 0$

Example: Determine the following limits

a) $\lim_{x\to\infty}\left(5+\frac{1}{x}\right)$

b) $\lim_{x\to-\infty}\frac{\pi\sqrt{3}}{x^2}$

Solution

a) $\lim_{x\to\infty}\left(5+\frac{1}{x}\right) = \lim_{x\to\infty} 5 + \lim_{x\to\infty}\frac{1}{x} = 5+0=5.$

b) $\lim_{x\to-\infty}\frac{\pi\sqrt{3}}{x^2} = \pi\sqrt{3}\lim_{x\to-\infty}\frac{1}{x^2} = \pi\sqrt{3}\cdot 0 = 0.$

4.8.4 Limits of Rational Functions as $x \to \pm\infty$

Consider the function

$$r(x) = \frac{P(x)}{Q(x)}$$

To find

$$\lim_{x \to \pm\infty} r(x)$$

Divide $P(x)$ and $Q(x)$ by the leading (highest) power of x in $Q(x)$.

Here are three cases for limits of rational functions as $x \to \pm\infty$:

1. If $deg(P(x)) < deg(Q(x))$, $\lim_{x \to \pm\infty} r(x) = 0$.
2. If $deg(P(x)) = deg(Q(x))$, $\lim_{x \to \pm\infty} r(x) = \frac{a_n}{b_n}$, where a_n and b_n are the leading coefficients of $P(x)$ and $Q(x)$ respectively.
3. If $deg(P(x)) > deg(Q(x))$, $\lim_{x \to \pm\infty} r(x) = \pm\infty$.

Example: Determine the following limits

a) $\lim_{x\to\infty} \frac{11x+2}{2x^3-1}$
b) $\lim_{x\to-\infty} -\frac{15x}{7x+4}$
c) $\lim_{x\to\infty} \frac{5x^2+8x-3}{3x^2+2}$
d) $\lim_{x\to\infty} \frac{-4x^3+7x}{2x^2-3x-10}$

Solution

a) $P(x) = 11x + 2$ and $Q(x) = 2x^3 - 1$. Thus $deg(P(x)) = 1 < 3 = deg(Q(x))$. So,

$$\lim_{x\to\infty} \frac{11x+2}{2x^3-1} = \lim_{x\to\infty} \frac{\frac{11}{x^2} + \frac{2}{x^3}}{2 - \frac{1}{x^3}}, \quad \frac{11}{x^2}, \frac{2}{x^3}, \frac{1}{x^3} \to 0 \text{ as } x \to \infty$$
$$= \frac{0}{2} = 0$$

b) $P(x) = 15x$ and $Q(x) = 7x + 4$. Thus $deg(P(x)) = 1 = deg(Q(x))$. So,

$$\lim_{x\to-\infty} -\frac{15x}{7x+4} = -\lim_{x\to-\infty} \frac{15}{7+\frac{4}{x}}, \quad \frac{4}{x} \to 0 \text{ as } x \to -\infty$$
$$= -\frac{15}{7}$$

c) $P(x) = 5x^2 + 8x - 3$ and $Q(x) = 3x^2 + 2$. Thus, $deg(P(x)) = 2 = deg(Q(x))$. So,

$$\lim_{x\to\infty} \frac{5x^2+8x-3}{3x^2+2} = \lim_{x\to\infty} \frac{5+\frac{8}{x}-\frac{3}{x^2}}{3+\frac{2}{x^2}}, \quad \frac{8}{x}, \frac{3}{x^2}, \frac{2}{x^2} \to 0 \text{ as } x \to \infty$$
$$= \frac{5}{3}$$

d) $P(x) = -4x^3 + 7x$ and $Q(x) = 2x^2 - 3x - 10$. Thus $deg(P(x)) = 3 > 2 = deg(Q(x))$. So,

$$
\begin{aligned}
\lim_{x\to\infty} \frac{-4x^3+7x}{2x^2-3x-10} &= \lim_{x\to\infty} \frac{-4x+\frac{7}{x}}{2-\frac{3}{x}-\frac{10}{x^2}} \\
&= \lim_{x\to\infty} \frac{-4x}{2} \\
&= -2 \lim_{x\to\infty} x \\
&= -2\cdot\infty = -\infty
\end{aligned}
$$

Theorem 4.8.5 The Sandwich (Squeeze) Theorem

Let $f(x)$, $g(x)$ and $h(x)$ be functions such that $f(x) \leq h(x) \leq g(x)$ for all x in the interval containing a (except possibly at a). If

$$\lim_{x\to a} f(x) = L = \lim_{x\to a} g(x)$$

then

$$\lim_{x\to a} h(x) = L$$

Example: Determine the following limits

a) $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right)$

b) $\lim_{x\to -\infty} \left(2 + \frac{\sin x}{x}\right)$

Solution

a) Since

$$-1 \leq \sin x \leq 1$$

then

$$-1 \leq \sin\left(\frac{1}{x^2}\right) \leq 1$$

Thus,

$$-x^2 \leq x^2 \sin\left(\frac{1}{x^2}\right) \leq x^2$$

and

$$\lim_{x\to 0}(-x^2) = 0 = \lim_{x\to 0} x^2$$

Therefore,

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right) = 0$$

b) We have that

$$\lim_{x\to -\infty}\left(2+\frac{\sin x}{x}\right) = \lim_{x\to -\infty} 2 + \lim_{x\to -\infty}\frac{\sin x}{x} = 2 + \lim_{x\to -\infty}\frac{\sin x}{x}$$

Since

$$-1 \le \sin x \le 1$$

then

$$-\frac{1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$$

So,

$$\lim_{x\to -\infty} -\frac{1}{x} = 0 = \lim_{x\to -\infty}\frac{1}{x}$$

Therefore

$$\lim_{x \to -\infty} \frac{\sin x}{x} = 0$$

$\therefore \lim_{x \to -\infty} \left(2 + \frac{\sin x}{x}\right) = 2 + 0 = 2$

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4.8.5 Continuous Functions

Definition 4.8.7 Continuity at a point

A function $y = f(x)$ is **continuous** at a point a on its domain if

$$\lim_{x \to a} f(x) = f(a) \tag{8}$$

Definition 4.8.7 implicitly requires three things if $f(x)$ is continuous at a:

1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exists
3. $\lim_{x \to a} f(x) = f(a)$

Definition 4.8.8 Continuity at an end-point

A function is **continuous from the right at** $x = a$ if

$$\lim_{x \to a^+} f(x) = f(a) \tag{9}$$

and is **continuous from the left at** $x = a$ if

$$\lim_{x \to a^-} f(x) = f(a) \tag{10}$$

Discontinuity at a point

If a function f is not continuous at a point $x = a$, we say that $f(x)$ is discontinuous at $x = a$ and we call $x = a$ a point of discontinuity of $f(x)$.

Example

a) Determine whether the function

$$f(x) = \begin{cases} 1 - x^2, & 0 < x \leq 1 \\ x, & x > 1 \end{cases}$$

is continuous at $x = 1$

b) Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

Solution

a) We have that

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1} (1 - x^2) = 1 - 1^2 = 0$$

and

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1} x = 1$$

and

$$f(1) = 1 - (1)^2 = 0$$

Since $\lim_{x\to 1^-} f(x) = 0 \neq 1 = \lim_{x\to 1^+} f(x)$, then $f(x)$ is not continuous at $x = 1$.

b) If $-1 < a < 1$, then

$$\begin{aligned}\lim_{x\to a} f(x) &= \lim_{x\to a}\left(1 - \sqrt{1-x^2}\right)\\ &= 1 - \lim_{x\to a}\sqrt{1-x^2}\\ &= 1 - \sqrt{1-a^2} = f(a)\end{aligned}$$

Therefore $f(x)$ is continuous at $x = a$ if $-1 < a < 1$.

Furthermore,

$$\begin{aligned}\lim_{x\to -1^+} f(x) &= \lim_{x\to -1}\left(1-\sqrt{1-x^2}\right)\\ &= 1-\lim_{x\to -1}\sqrt{1-x^2}\\ &= 1-\sqrt{1-(-1)^2} = f(-1)\end{aligned}$$

and

$$\begin{aligned}\lim_{x\to 1^-} f(x) &= \lim_{x\to 1}\left(1-\sqrt{1-x^2}\right)\\ &= 1-\lim_{x\to 1}\sqrt{1-x^2}\\ &= 1-\sqrt{1-1^2} = f(1)\end{aligned}$$

$\therefore\ f(x) = 1-\sqrt{1-x^2}$ is continuous on the interval $[-1,1]$.

Exercise 4.6

1. Consider the function

$$f(x) = \begin{cases} 1 - x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

   a) Graph the function $f(x)$.
   b) Find $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$.
   c) Does $\lim_{x\to 1} f(x)$ exist? If so, what is it. If not, why not?

2. Determine the following limits

   a) $\lim_{x\to 0} \frac{3}{1+\sqrt{3x+1}}$
   b) $\lim_{x\to 0} \frac{\sin 2x}{x}$
   c) $\lim_{x\to \pm\infty} \frac{10x^5+x^4+31}{x^6}$
   d) $\lim_{x\to \infty} \frac{2+\sqrt{x}}{2-\sqrt{x}}$
   e) $\lim_{x\to 2} \frac{3}{x-2}$
   f) $\lim_{x\to \infty} f(x)$ if $\frac{2x^2}{x^2+1} < f(x) < \frac{2x^2+5}{x^2}$

4. Find the value of a for which the function

$$f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x \neq -1 \\ a, & x = -1 \end{cases}$$

is continuous.

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Exponential and Logarithmic Functions

(1.) EXPONENTIAL FUNCTIONS

Definition 4.4.1

Consider a function of the form

$$f(x) = a^x, \qquad a > 0 \tag{11}$$

such a function is called an **exponential function**.

Let us consider three cases for the base a as follows:

1. If $a = 1$, then $f(x) = 1^x = 1$.
2. If $0 < a < 1$, then the graph of $f(x) = a^x$ has the following properties:

   I. $f(x)$ does not intersect the $x-$axis however, it intersect the $y-$axis at $y = f(0) = 1$.
   II. $f(x) > 0$ for all $x \in \mathbb{R}$. Thus, $D_f = \mathbb{R}$ and $R_f = (0. + \infty)$.
   III. As $x \to +\infty$, $f(x) = a^x \to 0$ and as $x \to -\infty$, $f(x) \to +\infty$.

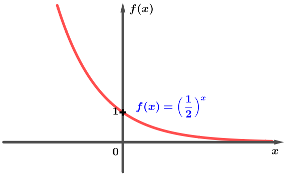

Figure 11: Graph of $f(x) = \left(\frac{1}{2}\right)^x$

For example, let $a = \frac{1}{2}$, the graph of $f(x) = \left(\frac{1}{2}\right)^x$ is shown in Figure 11 above.

3. If $a > 1$, then the graph of $f(x) = a^x$ has the following properties:
   1. $f(x)$ does not intersect the $x-$axis however, it intersect the $y-$axis at $y = f(0) = 1$.
   2. $f(x) > 0$ for all $x \in \mathbb{R}$. Since $a > 1$ then $a^x > 0$ for all $x \in D_f$. Thus, $D_f = \mathbb{R}$ and $R_f = (0. + \infty)$.
   3. As $x \to +\infty$, $f(x) = a^x \to +\infty$ and as $x \to -\infty$, $f(x) \to 0$.

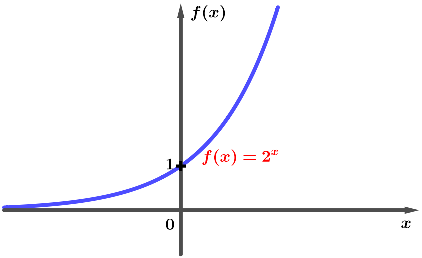

Figure 12: Graph of $f(x) = \left(\frac{1}{2}\right)^x$

For example, let $a = 2$, the graph of $f(x) = 2^x$ is shown in Figure 12 above.

In general, $f(x) = \left(\frac{1}{a}\right)^x = a^{-x}$ is a reflection of $g(x) = a^x$ on the $y-$axis.

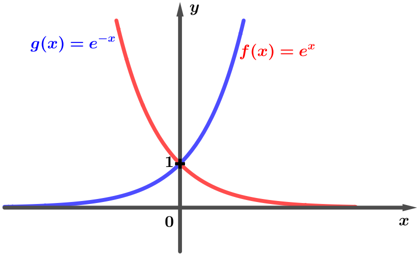

Figure 13: Graph of $f(x) = e^x$ and $g(x) = e^{-x}$

- A particular example of an exponential function is when $a = e = 2.718\ldots$.
- The function $f(x) = e^x$ is called the *natural exponential function*.
- Since $e > 1$ and $\frac{1}{e} < 1$, we can sketch the graphs of $f(x) = e^x$ and $g(x) = \left(\frac{1}{e}\right)^x = e^{-x}$ on the same axis as shown above.

(2.) LOGARITHMIC FUNCTIONS

Definition 4.4.2

A logarithmic function is a function of the form

$$f(x) = \log_a x \tag{12}$$

We consider the following cases for the base a

1. If $0 < a < 1$, then the graph of $f(x) = \log_a x$ has the following properties:
   1. $f(x)$ does not intersect the $y-$axis however, it intersect the $x-$axis at $x = 1$. Thus, $f(x) = \log_a x = 0 \implies x = a^0 = 1$.
   2. $f(x) = \log_a x$ is only defined for $x > 0$. Thus $D_f = (0, +\infty)$ and $R_f = \mathbb{R}$.

For example, let $a = \frac{1}{2}$, the graph of $f(x) = \log_{\frac{1}{2}} x$ is shown below.

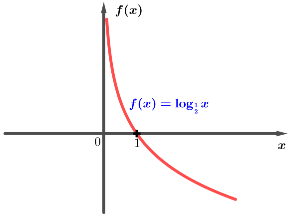

Figure 14: Graph of $f(x) = \log_{\frac{1}{2}} x$

2. If $a > 1$, then the graph of $f(x) = \log_a x$ has the following properties:

   1. $f(x)$ does not intersect the $y-$axis however, it intersect the $x-$axis at $x = 1$. Thus, $f(x) = \log_a x = 0 \implies x = a^0 = 1$.

   2. $f(x) = \log_a x$ is only defined for $x > 0$. Thus $D_f = (0, +\infty)$ and $R_f = \mathbb{R}$.

For example, let $a = 2$, the graph of $f(x) = \log_2 x$ is shown below.

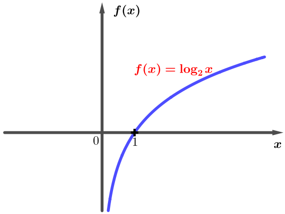

Figure 15: Graph of $f(x) = \log_2 x$

(3.) RELATIONSHIP BETWEEN EXPONENTIAL & LOGARITHMIC FUNCTIONS

Let's investigate the relationship between the exponential function $f(x) = e^x$ and logarithmic functions $g(x) = \log_e x = \ln x$ by looking at their graphs a shown in Figure ?? below.

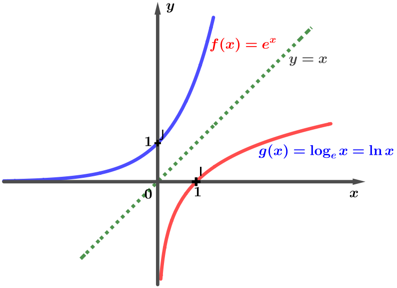

Figure 16: Graph of $f(x) = e^x$ and $g(x) = \log_e x = \ln x$

Similarly, for the exponential function $f(x) = \left(\frac{1}{2}\right)^x$ and the logarithmic function $g(x) = \log_{\frac{1}{2}} x$ their graphs are shown in Figure 17 below.

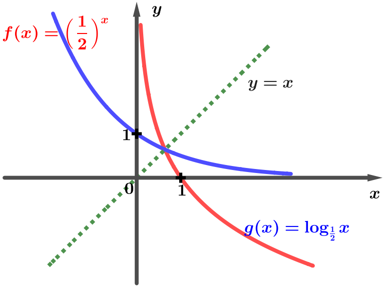

Figure 17: Graph of $f(x) = \left(\frac{1}{2}\right)^x$ and $g(x) = \log_{\frac{1}{2}} x$

Observations: From Figures ?? and 17 we observe that:

- logarithmic $g(x) = \log_a x$ is a reflection of the exponential function $f(x) = a^x$ on the line $y = x$.
- The logarithmic function $g(x) = \log_e x$ is the inverse of the exponential function $f(x) = e^x$ i.e. $g(x) = \log_e x = f^{-1}(x)$.
- Similarly, $g(x) = \log_{\frac{1}{2}} x$ is the inverse of $f(x) = \left(\frac{1}{2}\right)^x$ i.e. $g(x) = \log_{\frac{1}{2}} x = f^{-1}(x)$.

Exercise 4.2

Sketch the graphs of the following functions of the same axis

a. $f(x) = 3^x$
b. $g(x) = \left(\frac{1}{3}\right)^x$
c. $h(x) = \log_3 x$
d. $i(x) = \log_{\frac{1}{3}} x$

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Hyperbolic Functions

Definition 4.5.1

Hyperbolic functions, $\sinh x$, $\cosh x$, $\tanh x$, etc. are certain even and odd combinations of exponential functions e^x and e^{-x}.

The six main hyperbolic functions are given below:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}$$

$$\text{csch } x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \qquad \text{sech } x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

The notation of hyperbolic functions implies a close relation to trigonometric, $\sin x$, $\cos x$, $\tan x$, etc. However, the relationship is algebraic rather than geometric. Figures 18, 19 and 20 shows the graphs of $\sinh x$, $\cosh x$ and $\tanh x$.

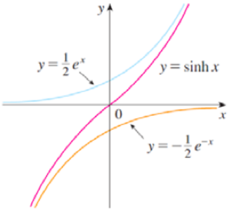

Figure 18: $\sinh x$

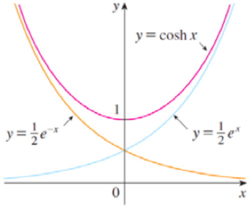

Figure 19: $\cosh x$

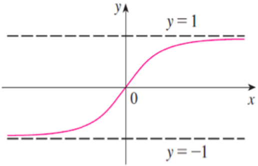

Figure 20: tanh

The graphs of hyperbolic sine and cosine functions can be sketched using graphical addition as shown in Figures 18 and 19.

Hyperbolic Identities

Similar trigonometric functions, hyperbolic functions satisfy a number of identities as shown below:

(1.) $\sinh(-x) = -\sinh(x)$ — $\sinh x$ *is odd*

(2.) $\cosh(-x) = \cosh(x)$ — $\cosh x$ *is even*

(3.) $\tanh(-x) = -\tanh(x)$ — $\tanh x$ *is odd*

(4.) $\cosh^2 x - \sinh^2 x = 1$

(5.) $1 - \tanh^2 x = \text{sech}^2 x$

(6.) $\coth^2 x - 1 = \text{csch}^2 x$

(7.) $\sinh(x \pm y) = \sinh x \cosh y \pm \sinh y \cosh x$

(8.) $\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

(9.) $\tanh(x \pm y) = \dfrac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$

Example: Solve the following exercises

1. Find the numerical value of each expression
   - i) $\sinh(\ln 2)$
   - ii) $\operatorname{sech}(0)$
   - iii) $\sinh(1)$
2. Prove that
   - i) $\sinh(-x) = -\sinh(x)$
   - ii) $\cosh x + \sinh x = e^x$
3. Prove that $\tanh(x+y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$

Solution:

1. i) $\sinh(\ln 2) = \frac{e^{\ln 2} - e^{-\ln 2}}{2} = \frac{2 - \frac{1}{2}}{2} = \frac{3}{4}$

   ii) $\text{sech}\,(0) = \frac{1}{\cosh(0)} = \frac{2}{e^{0} + e^{-0}} = 1$

   iii) $\sinh(1) = \frac{e^{1} - e^{-1}}{2} = \frac{e^{2} - 1}{2e}$

2. i) $\sinh(-x) = \frac{e^{-x} - e^{x}}{2} = -\frac{e^{x} - e^{-x}}{2} = -\sinh x$

   ii) $\cosh x + \sinh x = \frac{e^{x} + e^{-x}}{2} + \frac{e^{x} - e^{-x}}{2} = \frac{2e^{x}}{2} = e^{x}$

3. $\tanh(x+y) = \frac{\sinh(x+y)}{\cosh(x+y)}$

$$\begin{aligned}\frac{\sinh(x+y)}{\cosh(x+y)} &= \frac{\sinh x \cosh y + \cosh x \sinh y}{\cosh x \cosh y + \sinh x \sinh y} \\ &= \frac{\frac{\sinh x \cosh y}{\cosh x \cosh y} + \frac{\sinh y \cosh x}{\cosh x \cosh y}}{\frac{\sinh x \cosh y}{\cosh x \cosh y} + \frac{\sinh x \sinh y}{\cosh x \cosh y}} \\ &= \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}\end{aligned}$$

$\therefore \tanh(x+y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$.

Exercise 4.3

Show that

a) $\tanh(\ln x) = \frac{x^2-1}{x^2+1}$

b) $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$

The End of Chapter 1